

Final-Packaging — Standard Operating Procedure

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Audience: Maintenance Tech · Line Manager **Applies to:** Line-D, station 7, Final-Packaging

1. Purpose

Final-Packaging is the last station of Line-D. It wraps and boxes finished parts for shipment. A stoppage here blocks upstream Coating-Inspection within one buffer fill (≈ 10 seconds).

2. Normal operating envelope

- Ideal cycle time: **10 s**
- Max acceptable cycle time: **14 s**
- Expected reject rate: **0 %**
- Fault probability per cycle: **0.1 %**
- Input buffer capacity: **5 parts**

3. Idle reasons

idle_reason	Meaning here	First check
Starved	Coating-Inspection (station 6) is not feeding parts.	Verify Coating-Inspection state.
Blocked	Outbound shipping dock conveyor full.	Verify discharge conveyor.

4. Faults & corrective actions

4.1 Conveyor-Jam

- **Symptoms in telemetry:** fault_type = "Conveyor-Jam". Packaging halts; upstream blocks.
- **Likely root cause:** part misalignment, foreign object, or torn belt.
- **Corrective action:**
 1. LOTO. Clear the jam.
 2. Inspect belt for damage; replace if torn.

- **Restart criteria:** 20 cycles without slip.

4.2 Seal-Bar-Temp

- **Symptoms in telemetry:** fault_type = "Seal-Bar-Temp". Heat sealer out of temperature range; seal quality alarms.
- **Likely root cause:** heating element degraded; thermocouple drift.
- **Corrective action:**
 1. LOTO. Verify element resistance.
 2. Replace element or thermocouple.
- **Restart criteria:** stable seal temperature ± 3 °C.

4.3 Film-Feed-Error

- **Symptoms in telemetry:** fault_type = "Film-Feed-Error". Wrap film not advancing.
- **Likely root cause:** film roll empty, film tear, or feed roller slipping.
- **Corrective action:**
 1. Replace film roll if exhausted.
 2. Re-thread film; verify tension.
 3. Clean feed rollers.
- **Restart criteria:** continuous feed for 5 cycles.

5. Preventive maintenance schedule

- **Daily:** clean seal bar; check film stock.
- **Weekly:** belt tracking; clean feed rollers.
- **Monthly:** seal bar element check.
- **Quarterly:** full conveyor lubrication.

6. References

- Maintenance_Workflow_Overview.pdf
- Line-D_06_Coating-Inspection_SOP.pdf
- Production_Lines_Reference.pdf